

MULTI-AGENT SYSTEMS

AI Agent Collaboration in Fire & Safety Engineering

An educational guide to automating code analysis, drafting, & CAD exporting

| The Evolution: LLMs to Multi-Agents

Single Agent (Independent)

A single LLM equipped with tools (e.g., calculations or file parsing) acting alone. While powerful, attempting to solve both creative, spatial drawing design and rigorous code compliance in a single agent loop often leads to system fatigue, memory issues, or logical errors.

Multi-Agent System (Collaborative)

A structured network of multiple specialized agents. Each agent possesses unique tools, instructions, and constraints. One agent's verified output seamlessly initiates the next agent's workflow, mimicking a professional engineering office layout.

| Agent #1: Fire & Safety Engineer



1. Risk Assessment

Analyzes spatial properties to resolve **Occupancy & Fire Risk Classifications**, setting the structural rules for fire prevention.



2. Egress Planning

Performs automated calculations for maximum travel distances, stairs, and **Exit Route Planning** parameters to assure safe exits.



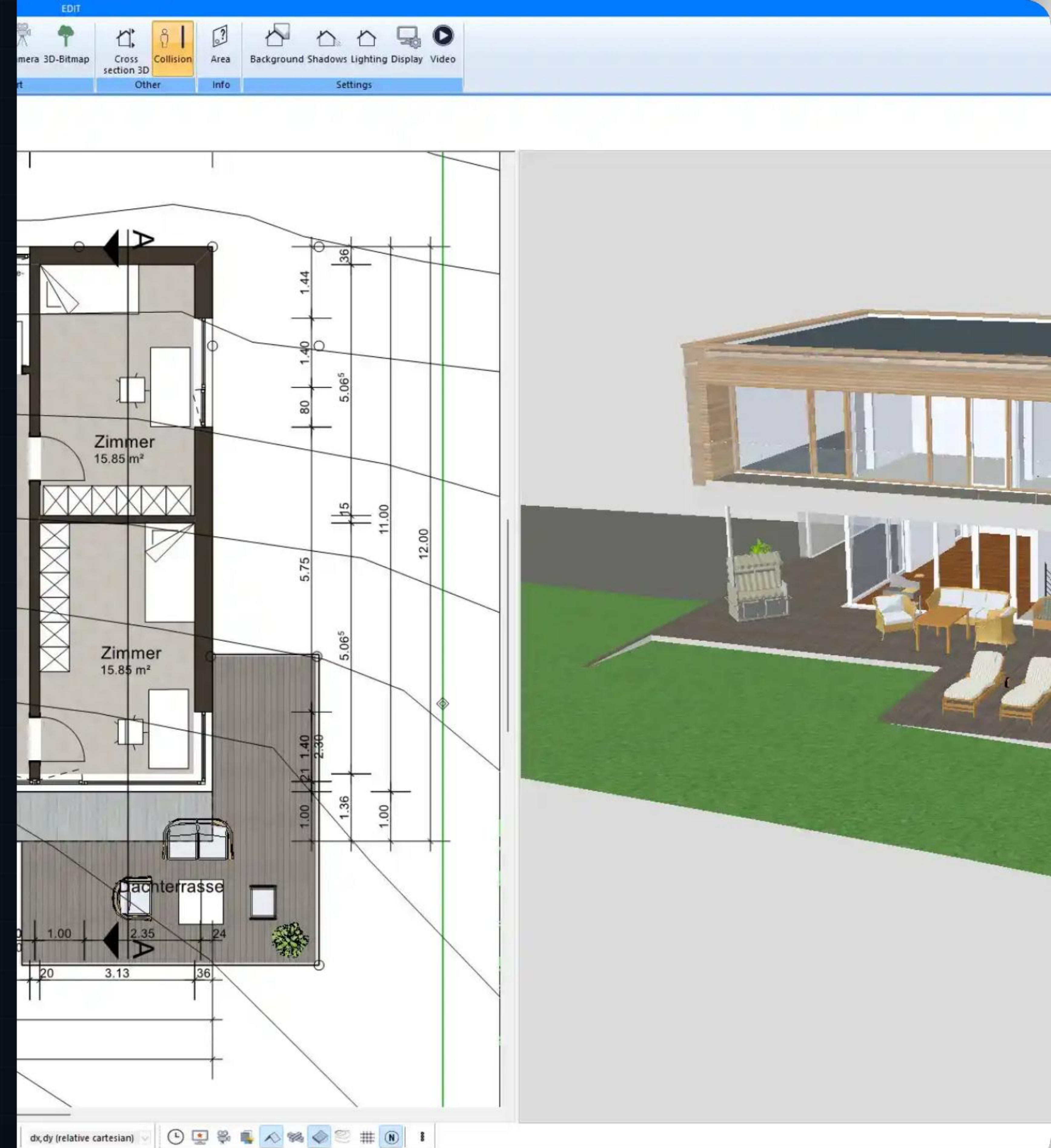
3. Sprinkler Checks

Checks compliance limits to map out critical fire pump zones, tank sizes, and runs rigorous **Sprinkler Requirement Checks**.

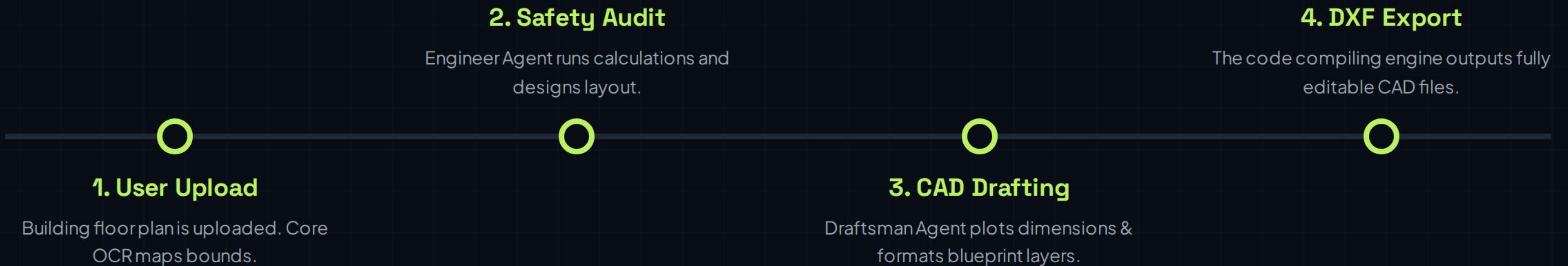
Agent #2: The CAD Draftsman

Visual & Spatial Translation

Directly implementing Agent 12 from `**image_4d6144.png**`, this agent translates calculated safety bounds into a spatial CAD canvas. It generates 2D Floor Plans, creates organized drawing layers, appends technical annotations, coordinates, and exports editable **DXF designs** directly.



| The Joint Pipeline: **Execution Loop**



Behind the Science: **Hydraulics Math**

$$p = \frac{4.52 \times Q^{1.85}}{C^{1.85} \times d^{4.87}}$$

Calculation Factor	Target (Light Hazard)	System Output	Automated Action taken by Agent
Pipe Roughness (C)	120 (Standard steel)	120 (Verified)	Set automatically via material lookup API
Discharge Flow (Q)	≥ 80 L/min per head	84.5 L/min (Pass)	Validated via virtual hydropressure simulator
Pipe Diameter (d)	Sized for friction loss	1.25 inches (Optimum)	Calculated dynamically to minimize budget costs

| Building Block: **Developing Agents**

-  **1. System Persona Definition:** Instruct your LLM with strict roles (e.g., *`"You are a draftsman compiling CAD files. Always format measurements in metric coordinates."`*).
-  **2. Integration of Native APIs:** Do not let the LLM hallucinate math. Build python tools containing geometric libraries, pressure equations, and DXF-parsing packages.
-  **3. Iterative Feedback Routing:** Program state logic constraints (such as LangGraph). The Draftsman Agent must not draft coordinates until the Engineer Agent passes validation checks.